INCH-POUND

MIL-R-55342F
28 June 1995
SUPERSEDING
MIL-R-55342E
26 January 1991

MILITARY SPECIFICATION

RESISTORS, FIXED, FILM, CHIP, ESTABLISHED RELIABILITY, GENERAL SPECIFICATION FOR

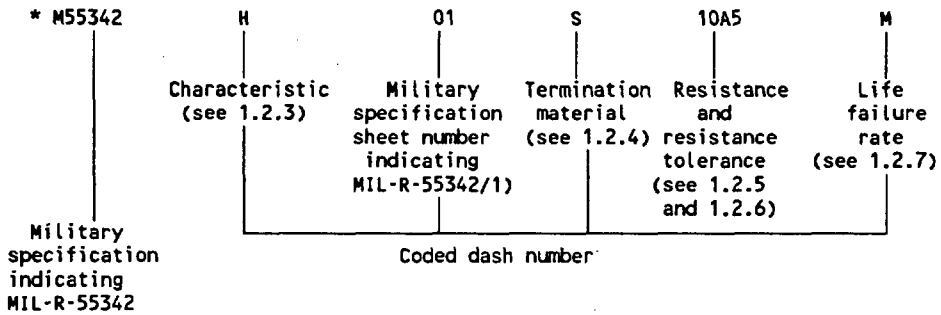
This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for fixed, film, chip resistors primarily intended for incorporation into hybrid microelectronic circuits and surface mounting applications. Resistors covered by this specification will have an established reliability with life failure rates ranging from 1 percent to 0.001 percent per 1,000 hours (see 1.2.7). These failure rates are established at 60 percent confidence on the basis of life tests. The failure rate, identified by the appropriate symbol, is referred to operation at full rated voltage and rated temperature, with a permissible change in resistance of ±2 percent as the criteria for failure. Table I provides a summary of performance characteristics for these resistors. A part per million (PPM) quality system is used for documenting and reporting the average outgoing quality of resistors supplied to this specification. Statistical process control (SPC) techniques are required in the manufacturing process to minimize variation in production of resistors supplied to the requirements of this specification.

1.2 Classification.

1.2.1 Part or Identifying Number (PIN). Resistors specified herein (see 3.1) will be identified by a PIN which will consist of the basic number of the military specification sheet and a dash number. Each military specification sheet covers a different resistor style. The dash number will be coded to provide information concerning the resistor characteristic, resistance value, resistance tolerance, termination material, and life failure rate. The PIN shall be in the following form with the dash number derived as indicated:



* A "D" prefix is used for metric detail specification sheets.

1.2.2 Style. The style (see 3.1) is identified by the two-letter symbol "RM" followed by a four digit number; the letters identify fixed, film resistors, in a flat chip configuration.

1.2.3 Characteristic. The characteristic is identified by the single letter E, H, K, or M, in accordance with table I.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Research Laboratory, ATTN: AMSRL-PS-DC, Fort Monmouth, NJ 07703-5601, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5905

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TABLE I. Characteristics.

Test or condition	Units	Symbol			
		E	H	K	M
Maximum ambient temperature at full wattage (see 3.6)	Temperature °C	70	70	70	70
Maximum ambient temperature at zero wattage (see 3.6)	Temperature °C	150	150	150	150
Thermal shock (see 3.9)	Maximum percent change in resistance (0.01 ohm additional allowed for measurement error)	±.1%	±.25%	±.5%	±.5%
Low temperature operation (see 3.10)		±.1%	±.25%	±.25%	±.5%
Short time overload (see 3.11)		±.1%	±.1%	±.25%	±.5%
High temperature exposure (see 3.12)		.1%	±.2%	±.5%	±1.0%
Resistance to bonding exposure (see 3.13)		±.2%	±.25%	±.25	±.25%
Moisture resistance (see 3.14)		±.2%	±.4%	±.5%	±.5%
Resistance temperature coefficient (see 3.15)	PPM/°C	±25	±50	±100	±300
Life (see 3.16) 2,000 hours FR level		±.5% ±2.0%	±.5% ±2.0%	±.5% ±2.0%	±2.0% ±2.0%
Resistance tolerance ± percent (see table V)	± percent	0.1 1.0 2.0	0.1 1.0 2.0 5.0	0.1 1.0 2.0 5.0 10.0	1.0 2.0 5.0 10.0

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1.2.4 Termination material. The termination material designation will be in accordance with table II.

TABLE II. Termination materials.

Type	Material	Termination area	Code letters
Wire bondable	Gold	One surface Bonding pads	W 3/ P
Solderable 1/	Base metallization barrier metal, solder coated	Wraparound 2/	B
	Pretinned	Wraparound 2/ One surface	R 4/ S 3/
Epoxy bondable	Gold	Wraparound 2/	G
	Platinum gold	Wraparound 2/ One surface	U T 3/
	Palladium/silver or Platinum/silver	Wraparound 2/	C
	Palladium/silver or Platinum/silver	One surface	D 3/

1/ Solderable terminations will be pretinned for solder reflow operation and will meet the solderability test. The pretinning will be, as a minimum, on at least the bottom and ends of the chip and only those surfaces must meet the solderability test.
2/ Wrap-around type will be illustrated on the detail specifications.
3/ See 6.4.4.
4/ Inactive for new design (see 6.7).

1.2.5 Resistance. The nominal resistance expressed in ohms is identified by four characters consisting of three digits and a letter. The letter is used simultaneously as a decimal point, multiplier, and a resistance tolerance designator. For resistance values:

a. Greater than or equal to 1 ohm but less than 1,000 ohms, the letters A, D, G, J, and M are used to represent a decimal point, depending on the resistance tolerance (see table III).

b. Greater than or equal to 1,000 ohms but less than 1 megohm, the letters B, E, H, K, and N are used to represent a decimal point, depending on the resistance tolerance (see table III).

c. Greater than or equal to 1 megohm, the letters C, F, T, L, and P are used to represent a decimal point, depending upon the resistance tolerance (see table III). All digits preceding and following the letters (A, B, C, D, E, F, G, H, T, L, J, K, M, N, and P) of the group represent significant figures. The resistance value designators are shown in table III. Minimum and maximum resistance values will be as specified herein (see 3.1). The standard values for every decade will follow the sequence specified in table IV of MIL-R-55342 for resistance tolerances 1.0, 2.0, 5.0, and 10.0 percent. The resistance values for tolerance 0.1 percent maybe any value within the limits specified (see 3.1), but it is preferred that values be chosen from the 0.1 percent column of table IV of MIL-R-55342.

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TABLE III. Designator of resistance values for resistance tolerances.

Designator for 0.1 percent tolerance	Resistance ohms
1A00 to 9A88 inclusive	1.00 to 9.88 inclusive
10A0 to 98A8 inclusive	10.0 to 98.8 inclusive
100A to 988A inclusive	100 to 988 inclusive
1B00 to 9B88 inclusive	1,000 to 9,880 inclusive
10B0 to 98B8 inclusive	10,000 to 98,800 inclusive
100B to 988B inclusive	100,000 to 988,000 inclusive
1C00 to 9C88 inclusive	1,000,000 to 9,880,000 inclusive
10C0	10,000,000
Designator for 1 percent tolerance	**Resistance ohms**
1D00 to 9D76 inclusive	1.00 to 9.76 inclusive
10D0 to 97D6 inclusive	10.0 to 97.6 inclusive
100D to 976D inclusive	100 to 976 inclusive
1E00 to 9E76 inclusive	1,000 to 9,760 inclusive
10E0 to 97E6 inclusive	10,000 to 97,600 inclusive
100E to 976E inclusive	100,000 to 976,000 inclusive
1F00 to 9F76 inclusive	1,000,000 to 9,760,000 inclusive
10F	10,000,000
Designator 2 percent tolerance	**Resistance ohms**
1G00 to 9G10 inclusive	1.00 to 9.10 inclusive
10G0 to 91G0 inclusive	10.0 to 91.0 inclusive
100G to 910G inclusive	100 to 910 inclusive
1H00 to 9H10 inclusive	1,000 to 9,100 inclusive
10H0 to 91H0 inclusive	10,000 to 91,000 inclusive
100H to 910H inclusive	100,000 to 910,000 inclusive
1T00 to 9T10 inclusive	1,000,000 to 9,100,000 inclusive
10T0	10,000,000
Designator for 5 percent tolerance	**Resistance ohms**
1J00 to 9J10 inclusive	1.00 to 9.10 inclusive
10J0 to 91J0 inclusive	10.0 to 91.0 inclusive
100J to 910J inclusive	100 to 910 inclusive
1K00 to 9K10 inclusive	1,000 to 9,100 inclusive
10K0 to 91K0 inclusive	10,000 to 91,000 inclusive
100K to 910K inclusive	100,000 to 910,000 inclusive
1L00 to 9L10 inclusive	1,000,000 to 9,100,000 inclusive
10L0	10,000,000
Designator for 10 percent tolerance	**Resistance ohms**
1M00 to 9M10 inclusive	1.00 to 8.20 inclusive
10M0 to 91M0 inclusive	10.0 to 82.0 inclusive
100M to 910M inclusive	100 to 820 inclusive
1N00 to 9N10 inclusive	1,000 to 8,200 inclusive
10N0 to 91N0 inclusive	10,000 to 82,000 inclusive
100N to 910N inclusive	100,000 to 820,000 inclusive
1P00 to 9P10 inclusive	1,000,000 to 8,200,000 inclusive
10P0	10,000,000

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TABLE IV. Standard resistance values for the 10 to 100 decade.

0.1	1.0	2.0 5.0	10.0	0.1	1.0	2.0 5.0	10.0	0.1	1.0	2.0 5.0	10.0
10.00	10.00	10.00	10.00	20.00	20.00	20.00		37.90			
10.10				20.30				38.30	38.30		
10.20	10.20			20.50	20.50			38.80			
10.40				20.80						39.00	39.00
10.50	10.50			21.00	21.00			39.20	39.20		
10.60				21.30				39.70			
10.70	10.70			21.50	21.50			40.20	40.20		
10.90				21.80				40.70			
11.00	11.00	11.00				22.00	22.00	41.20	41.20		
11.10				22.10	22.10			41.70			
11.30	11.30			22.30				42.20	42.20		
11.40				22.60	22.60			42.70			
11.50	11.50			22.90						43.00	
11.70				23.20	23.20			43.20	43.20		
11.80	11.80			23.40				43.70			
12.00		12.00	12.00	23.70	23.70			44.20	44.20		
12.10	12.10			24.00		24.00		44.80			
12.30				24.30	24.30			45.30	45.30		
12.40	12.40			24.60				45.90			
12.60				24.90	24.90			46.40	46.40		
12.70	12.70			25.20				47.00		47.00	47.00
12.90				25.50	25.50			47.50	47.50		
13.00	13.00	13.00		25.80				48.10			
13.20				26.10	26.10			48.70	48.70		
13.30	13.30			26.40				49.30			
13.50				26.70	26.70			49.90	49.90		
13.70	13.70					27.00	27.00	50.50			
13.80				27.10						51.00	51.00
14.00	14.00			27.40	27.40			51.10	51.10		
14.20				27.70				51.70			
14.30	14.30			28.00	28.00			52.30	52.30		
14.50				28.40				53.00			
14.70	14.70			28.70	28.70			53.60	53.60		
14.90				29.10				54.20			
15.00	15.00	15.00	15.00	29.40	29.40			54.90	54.90		
15.20				29.80				55.60			
15.40	15.40					30.00				56.00	56.00
15.60				30.10	30.10			56.20	56.20		
15.80	15.80			30.50				56.90			
16.00		16.00		30.90	30.90			57.60	57.60		
16.20	16.20			31.20				58.30			
16.40				31.60	31.60			59.00	59.00		
16.50	16.50			32.00				59.70			
16.70				32.40	32.40			60.40	60.40		
16.90	16.90			32.80				61.20			
17.20						33.00	33.00	61.90	61.90		
17.40	17.40			33.20	33.20					62.00	
17.60				33.60				62.60			
17.80	17.80			34.00	34.00			63.40	63.40		
18.00		18.00	18.00	34.40				64.20			
18.20	18.20			34.80	34.80			64.90	64.90		
18.40				35.20				65.70			
18.70	18.70			35.70	35.70			66.50	66.50		
18.90						36.00		67.30			
19.10	19.10			36.10						68.00	68.00
19.30				36.50	36.50			68.10	68.10		
19.60	19.60			37.00				69.00			
19.80				37.40	37.40						

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TABLE IV. Standard resistance values for the 10 to 100 decade - Continued.

0.1	1.0	2.0 5.0	10.0	0.1	1.0	2.0 5.0	10.0	0.1	1.0	2.0 5.0	10.0
69.80	69.80			79.60				88.70	88.70		
70.60				80.60	80.60			89.80			
71.50	71.50			81.60				90.90	90.90		
72.30						82.00	82.00			91.00	
73.20	73.20			82.50	82.50			92.00			
74.10				83.50				93.10	93.10		
75.00	75.00	75.00	75.00	84.50	84.50			94.20			
75.90				85.60				95.30	95.30		
76.80	76.80			86.60	86.60			96.50			
77.70				87.60				97.60	97.60		
78.70	78.70							98.80			

1.2.6 Resistance tolerance and multipliers. The resistance tolerance and multipliers are identified by a single letter in accordance with table V.

TABLE V. Resistance tolerance and multipliers.

Resistance tolerance (±)	Multiplier	Symbol
0.1	X1	A
0.1	X1,000	B
0.1	X1,000,000	C
1.0	X1	D
1.0	X1,000	E
1.0	X1,000,000	F
2.0	X1	G
2.0	X1,000	H
2.0	X1,000,000	T
5.0	X1	J
5.0	X1,000	K
5.0	X1,000,000	L
10.0	X1	M
10.0	X1,000	N
10.0	X1,000,000	P

1.2.7 Life failure rate designation. The life failure rate designation as shown in table VI is signified by a single letter (M, P, R, S), which identifies the life failure rate level for which the resistor is qualified (see 4.5 and 6.3).

TABLE VI. Life failure rate level (established at 60 percent confidence).

Failure rate level designation	Failure rate percent/1,000 hours
M	1.0
P	0.1
R	0.01
S	0.001

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

MMM-A-1931 - Adhesive, Epoxy, Silver Filled, Conductive.

MILITARY

MIL-R-39032 - Resistors, Packaging of.
MIL-I-46058 - Insulating, Compound, Electrical (For coating printed circuit assembles).

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-690 - Failure Rate Sampling Plans and Procedures.
MIL-STD-790 - Product Assurance Program for Electronic and Fiber Optic Parts.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(See supplement 1 for list of associated detail specifications.)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA-554 - Assessment of outgoing nonconforming levels in Parts Per Million (PPM). (DoD adopted).
EIA-557 - Statistical Process Control Systems (DoD adopted).

(Applications for copies should be addressed to Electronic Industries Association (EIA), 2001 Pennsylvania Avenue NW, Washington DC 20006.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B545 - Standard Specification for Electrodeposited Coating of Tin.

(Application for copies should be addressed to the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Detail requirements. The individual item requirements shall be as specified herein and in accordance with the applicable detail specification. In the event of any conflict between the requirements of this specification and the detail specifications, the latter shall govern (see 6.2).

3.2 Qualification. Resistors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list (QPL). In addition, the manufacturer shall obtain certification from the qualifying activity that the reliability assurance requirements of 4.1.2 have been met and are being maintained. Authorized distributors that are approved to MIL-STD-790 distributor requirements by the QPL manufacturer are listed in the QPL.

3.3 Reliability and quality.

3.3.1 Reliability. Reliability of resistors furnished under this specification shall be established and maintained in accordance with the requirements and procedures specified in MIL-STD-790 and MIL-STD-690 with details and exceptions specified in 4.1.2, 4.4.4, 4.5, and 4.6.2.1.1.

3.3.2 Quality.

3.3.2.1 Statistical process control (SPC). The contractor shall implement and use statistical process control techniques in the manufacturing process for parts covered by this specification. The SPC program shall be developed and maintained in accordance with EIA-557. The SPC program shall be documented and maintained as part of the overall reliability assurance program as specified in MIL-STD-790. For manufacturers initially qualifying to this specification the implementation date for SPC shall be 6 months from the date of initial qualification. In addition, the manufacturer shall demonstrate resistance temperature characteristic (RTC) control in the process. Processes for application of SPC techniques should include but are not limited to:

a. Formation of resistance elements.

b. Trimming.

c. Termination finish.

3.3.2.2 Quality levels. The quality of lots that have been subject to and passed the subgroup 1 100 percent screening inspection of the group A inspection shall be established and maintained in accordance with 4.6.1.2.3 and EIA-554 method B. Individual PPM defect level (i.e., PPM-2 and PPM-3) and an overall PPM defect level (i.e., PPM-5) shall be established based on the tests prescribed in the subgroup 3 test of the group A inspections. The defect level for PPM-2 shall be less than 100 PPM. Data shall not be excluded from the appropriate PPM calculation unless specifically authorized by the qualifying activity. Guidance for the exclusion of data is specified in EIA-554. For manufacturers initially qualifying to this specification the implementation date for PPM verification shall be 6 months from the date of initial qualification.

3.3.2.2.1 Noncompliance. The contractor shall notify the qualifying activity when the 100 PPM level is reached or exceeded for PPM-2. The contractor shall provide sufficient information to the qualifying activity documenting the causes of the problem and what corrective action is being taken. Failure to correct this problem shall be the basis for removal of the affected product from the QPL.

3.4 Materials. The materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the resistors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.5 Design, construction, and physical dimensions. Resistors shall be of the design, construction, and physical dimensions specified (see 3.1). Each resistor shall consist of a film type resistance element connected to metallized contact areas of terminations which form the electrical contacts of the resistor.

3.5.1 Passivation. Unless passivation is inherent in the device, the construction shall be passivated with a coating over the resistance element to protect the element from direct exposure to environment or handling contamination. Does not apply to trim cut.

3.5.2 Termination. Connection to the resistor element shall be made via metallized termination areas. Unless otherwise specified, both terminations (areas) shall be available on one surface of the body of the resistor chip (see 1.2.4).

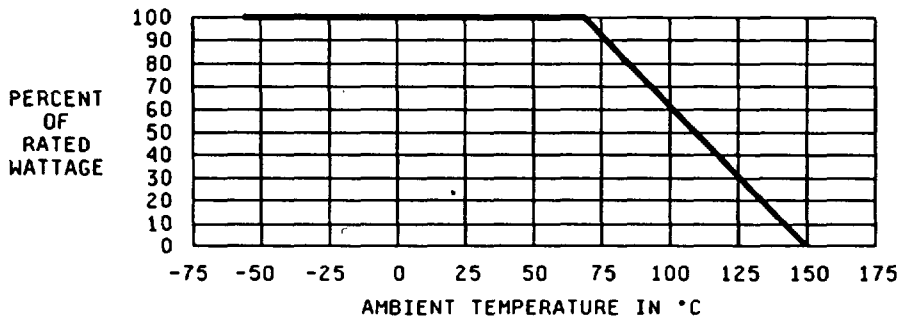
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3.5.2.1 Barrier metallization. The barrier metallization for the B termination (base metallization barrier metal, solder coated) shall be nickel. The metallization shall be a minimum of 50 microinches.

3.5.2.2 Tin plated finishes. Use of tin plating is prohibited as a final finish and as an undercoat (see 6.4.5). Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent.

3.5.3 Substrate bonding. The underside of the chip resistor (the surface opposite the side containing the resistance element) may be metallized, when specified, to provide for eutectic bonding of the chip to a substrate.

3.6 Power rating. The resistors shall have a power rating based on continuous full-load operation at an ambient temperature of 70°C. For operation at temperatures in excess of 70°C the load shall be derated in accordance with figure 1.



NOTE: This curve indicates the percentage of nominal wattage to be applied at temperature higher than 70°C. This curve applies only to units mounted on a board (see 4.7.1.3); however, at no time shall the applied voltage exceed the maximum for each style (see 3.1).

FIGURE 1. Derating curve for high ambient temperatures.

3.7 Voltage rating. Resistors shall have a rated dc continuous working voltage or an approximate sinewave root-mean-square (rms) continuous working voltage corresponding to the wattage (power) rating, as determined from the following formula:

$$E=\sqrt{PR}$$

Where

E = Continuous rated dc or rms working voltage in volts
P = Rated wattage in watts
R = Nominal resistance in ohms

In no case shall the rated voltage be greater than the applicable maximum voltage (see 3.1).

3.8 DC resistance. When resistors are tested as specified in 4.7.2, the dc resistance shall be within the specified tolerance of the nominal resistance (see 1.2.5).

3.9 Thermal shock. When resistors are tested as specified in 4.7.3, there shall be no evidence of mechanical damage; the change in resistance shall be within the specified initial resistance tolerance of the nominal resistance for group A tests and the change in resistance for group C tests shall not exceed the value specified in table I.

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3.10 Low temperature operation. When resistors are tested as specified in 4.7.4, there shall be no evidence of mechanical damage; the change in resistance between the initial and the final measurement at 25°C ±5°C, shall not exceed the value specified in table I.

3.11 Short time overload. When resistors are tested as specified in 4.7.5, there shall be no evidence of arcing, burning, or charring, the change in resistance shall not exceed the value specified in table I.

3.12 High temperature exposure. When resistors are tested as specified in 4.7.6, there shall be no evidence of mechanical damage and the change in resistance shall not exceed the value specified in table I.

3.13 Resistance to bonding exposure. When resistors are tested as specified in 4.7.7, there shall be no evidence of mechanical damage, no demetallization or leaching, and the change in resistance shall not exceed the value specified in table I.

3.14 Moisture resistance. When resistors are tested as specified in 4.7.8, there shall be no evidence of mechanical damage; the change in resistance between the initial and final measurements shall not exceed the value specified in table I.

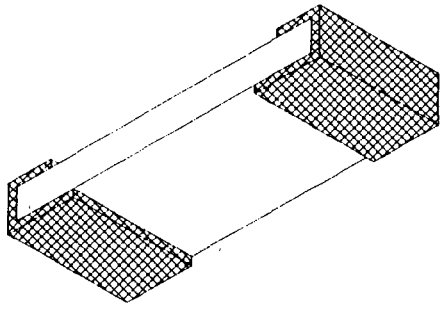
3.15 Resistance temperature characteristic. When resistors are tested as specified in 4.7.9, the resistance temperature characteristic, at each of the temperatures specified in table XIV referred to room ambient temperature, shall not exceed the values specified in table I.

3.16 Life.

3.16.1 Qualification. When resistors are tested as specified in 4.7.10, there shall be no evidence of mechanical damage. The change in resistance between the initial measurement and any succeeding measurements up to and including 2,000 hours, shall not exceed the value specified in table I.

3.16.2 Failure rate level determination. When resistors are tested as specified in 4.7.10, there shall be no evidence of mechanical damage to the resistance element, or enclosure. The change in resistance between the initial measurement and any of the succeeding measurements shall not exceed the value specified in table I. This single failure criteria shall be applicable to all measurements during the life test for purposes of determining failure rate level qualification and is applicable as a parallel requirement with 3.16.1 to measurements made during the life test specified for qualification inspection.

3.17 Solderability (applicable to terminations B, R, and S). When resistors are tested as specified in 4.7.11, the immersed metallized surface shall be at least 95 percent covered with a new clean smooth coating and shall exhibit no demetallization or leaching of the terminal areas. The remaining 5 percent may contain only small pinholes or rough spots, these shall not be concentrated in one area. In case of dispute, the percentage of coverage with pinholes or rough spots shall be determined by actual measurement of these areas, as compared to the total area. The top surface of the solder pad need not be considered in the evaluation for solderability (see figure 2).



NOTES:
1. Top solder pad need not be evaluated for solderability.
2. Resistive element on top.
3. Ends and bottom are surfaces to be evaluated.

FIGURE 2. Solderability coverage.

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3.18 Mounting integrity.

3.18.1 Solder mounting integrity (applicable to terminations B, R, and S). When resistors are tested as specified in 4.7.12.1, there shall be no evidence of mechanical damage.

3.18.2 Bondable mounting integrity (applicable to terminations C, D, G, T, and U). When resistors are tested as specified in 4.7.12.2, there shall be no evidence of mechanical damage.

3.18.3 Wirebonding integrity (applicable to terminations W and P). When resistors are tested as specified in 4.7.12.3, there shall be no evidence of mechanical damage.

3.19 Resistance to solvents (applicable to resistors that are marked with color dots or screen printed). When resistors are tested as specified in 4.7.13, there shall be no evidence of mechanical damage and the marking shall remain legible. The color dot marking shall be in accordance with MIL-STD-1285.

3.20 Marking legibility.

3.20.1 Marking legibility test (applicable to laser marked resistors). When resistors are tested as specified in 4.7.14, the marking shall remain legible. The following is an example of minimum marking:

10A5 - Resistance, resistance tolerance, and multiplier

3.21 Hazardous material. Hazardous items are substances, mixtures, materials, components, or equipments that may cause personal injury, property damage, or environmental deterioration through transportation, use, or disposal. These items shall be marked in accordance with the requirements of public law and regulations.

3.22 Visual inspection. Resistors shall be inspected as specified in 4.7.1, to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.4, 3.5, 3.5.1 to 3.5.3 inclusive, 3.22.1 to 3.22.5d inclusive, 3.24, and figures 3 through 14).

3.22.1 Conductor metallization defects. Any resistor which exhibits the following defects in the active circuit metallization shall not be acceptable. The active circuit metallization is all metal or any other material used for interconnection except metallized scribe lines, unused bonding pads, and identification marking.

3.22.1.1 Conductor metallization scratches. A scratch is defined as any tearing defect that disturbs the original surface of the metallization.

a. Any scratch in metallization through which the underlying resistor material also appears to be scratched.

b. Any scratch in the interconnecting metallization which exposes resistive material, substrate, or oxide anywhere along its length and reduces the width of the connecting strip to less than 50 percent of its original width.

3.22.1.2 Conductor metallization voids. A void is defined as any region in the interconnecting metallization where the underlying resistive material or substrate is visible which is not caused by a scratch. Any void in the interconnecting metallization which leaves less than 50 percent of the original width undisturbed (see figure 3).

3.22.1.3 Conductor metallization nonadherence. Conductor metallization nonadherence is defined as any evidence of metallization lifting, peeling, or blistering (see figure 7).

3.22.1.4 Conductor metallization probing. Conductor metallization probing is defined as probe marks on the interconnecting metallization other than the bonding pads that violate the scratch or void criteria.

3.22.1.5 Conductor metallization bridging (see figure 6). Conductor metallization bridging defects are defined as follows:

a. Bridged metallization defect that reduces the distance between any two metallization areas to less than 50 percent or 0.3 mils, whichever is greater.

b. Bridging between metallization and resistor pattern not intended by design that reduces the distance between the two to less than 0.1 mils.

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3.22.1.6 Conductor metallization missing. Conductor metallization defects are defined as follows:

a. More than 5 percent of the metallization area is missing from the bonding pad (see figure 3).

b. Any shorting bar or metal pad that is missing (see figures 3 and 10).

3.22.1.7 Conductor metallization alignment. Conductor metallization alignment is defined as any misalignment between resistor and metallization such that more than 0.5 mil of resistor on a side is exposed (see figure 4).

3.22.1.8 Conductor metallization corrosion. Metallization corrosion is defined as any evidence of localized heavy stains, metallization corrosion, discoloration or mottled metallization (see figure 9).

3.22.1.9 Excessive metallization. Excessive buildup of termination on any surface material greater than .003 inch (0.08 mm) and .008 inch (0.20 mm) for solderable and bondable termination respectively (see figure 6).

3.22.2 Resistor defects. Any resistor which exhibits the following defects in the active area shall not be acceptable. The active area of a resistor is that part of a resistor pattern which remains in series connection between resistor terminals and is not shorted by metallization and the resistor element is not more than two rungs past laser trimmed link.

3.22.2.1 Resistor scratches (see figure 5). Resistor scratches are defined as follows:

a. Any scratch within the active resistor area.

b. A scratch will normally be of considerable length, 1 mil or longer and on fine lines, will extend continuously on adjacent lines. A scratch must be indicated by either disturbance of the surface resistor or extraneous, shiny material remaining from the instrument causing the scratch.

3.22.2.2 Resistor voids (see figures 3 and 11).

a. Any void or neckdown in the active resistor area which reduces the width of the area by more than 50 percent of the original area.

b. Any void in the resistor element at the termination.

3.22.2.3 Resistor nonadherence. Resistor nonadherence is defined as any evidence of resistor film lifting, peeling, or blistering (see figure 7).

3.22.2.4 Probe marks. Probe marks are defined as any probe mark on the active resistor area.

3.22.2.5 Resistor material corrosion. Resistor material corrosion is defined as follows:

a. Any evidence of localized heavy stains or corrosion of resistor material in the active resistor area (see figure 9).

b. Discoloration of tantalum based resistors due to thermal stabilization is not a cause for rejection.

3.22.2.6 Resistor bridging defects. Resistor bridging defects are defined as follows:

a. Resistor bridging caused by photolithographic defect that forms a conductive bridge across two or more adjacent active resistance areas where width of bridge path is less than one half the smallest line width in the pattern.

b. Partial bridging defect that reduces the distance between adjacent active resistance areas to less than 0.1 mil or 50 percent of the design separation, whichever is less, when caused by smears, photolithographic defects or other causes except that for a partial bridge within the lines and spacing of 0.1 mil design width, visual separation (evidence at 400X max) is sufficient for acceptance, and that apparent bridging when caused by laser trim slag in a nonlaser trim area is acceptable as long as the resistor material under the apparent bridge is passivated.

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3.22.2.7 Laser trim faults. Any resistor which exhibits the following defects in the active area shall not be acceptable.

a. Uncut material remaining after a laser cut due to "skipping" of laser beam (see figure 8).

b. If laser cut is not straight lined, the narrowest remaining width must be equal to or greater than the width of the narrowest line within the same resistor pattern.

c. Oxide voids, cracking, or similar damage caused to the underlying material by laser beam where such damage touches active interconnects, resistor area, or both.

d. For a resistor pattern with links designed for incremental laser trim, where edge of cut touches the active resistor area.

e. Any "scorching" of resistor areas by direct laser beam or spurious reflections caused by optics of the system.

f. Discoloration of tantalum based resistors in and around laser kerf is not cause for rejection.

g. For a cut in the resistor pattern which reduces a link or ladder width by more than 50 percent of the original width.

h. Usable resistance width is less than 50 percent of active resistor element (figure 11).

3.22.3 Oxide or substrate void. Oxide or substrate voids are defined as follows:

a. Any resistor which exhibits an oxide or substrate void shall not be acceptable.

b. Any oxide or substrate void is a fault in the oxide or substrate as evidenced by localized double or triple colored fringes at the edges of the defect at 100X max.

c. Oxide or substrate void that allowed bridging between any two resistors, metal areas or both, not intended by design.

d. Any evidence of oxide or substrate void under the metallization or resistor geometry.

3.22.4 Scribing and substrate defects. The following defects shall be cause for rejection:

a. Less than 0.5 mil oxide visible between active metallization and edge of substrate. Excluded from this are any inactive metallization lines (if applicable).

b. Any chip or crack in the active resistor.

c. Any crack that exceeds 3.0 mil in length or comes closer than 1.0 mil to an active resistor or bonding pad area on the substrate (see figure 12).

d. Any crack in a substrate that exceeds 1.0 mil in length and points towards the active resistor area.

e. A substrate, when scribed and broken, is left with an attached portion of an adjacent substrate which contains metallization or resistor material.

f. Any crack not originating at a substrate edge is cause for rejection (see figure 12).

3.22.5 Foreign material. Conductive foreign material is defined as any substance which appears opaque under all conditions of lighting and magnification used in routine visual inspection. Material is considered attached when it cannot be removed with a nominal gas blow (of 20 psig) of dry nitrogen or air.

a. Attached conductive foreign material which reduces the separation between metallization areas to less than 0.1 mil is cause for rejection.

b. Attached conductive foreign material which reduces the separation between metallization and resistor areas to less than 0.1 mil is cause for rejection.

c. More than seven areas having attached nonconductive or conductive foreign material that are not rejectable by the criteria of 3.22.5a or 3.22.5b shall be cause for rejection.

d. Ink residue on chip from the tester shall be cause for rejection.

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3.23 Marking. Marking of individual chip resistors is not required but is permitted. A noncorrosive label containing the following information shall be applied to each unit package. PIN, "JAN" marking, date code, source code, and manufacturer's production lot code (see 4.6.1.1.2). Date and source code shall be in accordance with MIL-STD-1285. The following is an example of the complete marking:

12345 - CAGE
7343J - Date code and JAN marking
MXXXXXH01 - Part number
S10A5M - Part number
- Manufacturer's production lot code

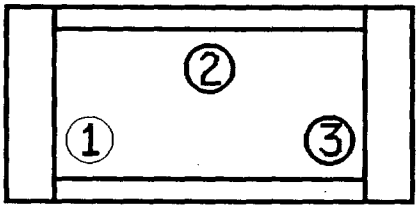
The date code shall be the date of the final assembly operation for the production lot as defined in 4.6.1.1.2. The common manufacturing record shall include the same date code as that placed on the parts covered by the record. At the option of the manufacturer, resistors may be marked (i.e., laser, color dot, screen printed, etc.). The marking process must be approved by the qualifying activity based on testing specified herein. The marking shall remain legible after all tests.

3.23.1 Marking (color dot, screen print, etc.).

3.23.1.1 Three dot option (designates 2, 5, and 10 percent values). An example of three dot marking option shall be as in the following example:

1st dot - 1st significant digit.
2nd dot - 2nd significant digit.
3rd dot - multiplier. (Ref. MIL-STD-1285)

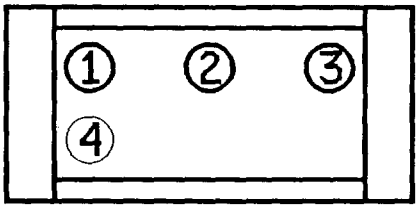
ex. - 330k ohm orange orange yellow



3.23.1.2 Four dot option (designates 1 percent values). An example of four dot marking option shall be as in the following example:

1st dot - 1st significant digit.
2nd dot - 2nd significant digit.
3rd dot - 3rd significant digit.
4th dot - multiplier. (Ref. MIL-STD-1285)

ex. - 30.1k ohm orange - black - brown red



3.23.2 Laser marking/screen printing/stamping. An example of the code used for laser marking, screen printing, or stamping shall be as shown in the following example.

330k ohms shall be coded 334 (three digit code)
12 ohms shall be coded 120 (three digit code)
30.1k ohms shall be coded 3012 (four digit code)
18.2 ohms shall be coded 18R2 (four digit code)

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3.23.3 JAN and J marking. The United States Government has adopted, and is exercising legitimate control over, the certification marks "JAN" and "J", respectively, to indicate electrical equipment, namely resistors, capacitors, electron tubes and the like, acquired by, or manufactured for use by, or for the Government in accordance with standard government specifications. Accordingly, resistors acquired to, and meeting all of the criteria specified herein and in applicable detail specification shall bear the certification mark "JAN", except that resistors too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number. Resistors furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein and in applicable detail specifications shall not bear "JAN" or "J". In the event a resistor sample fails to meet the requirements of this specification and the applicable detail specification, the manufacturer shall remove the "JAN" or "J" from the sample tested and also from all resistors represented by the sample. The United States Government has obtained certificate of registration no. 504,860 for the certification mark "JAN".

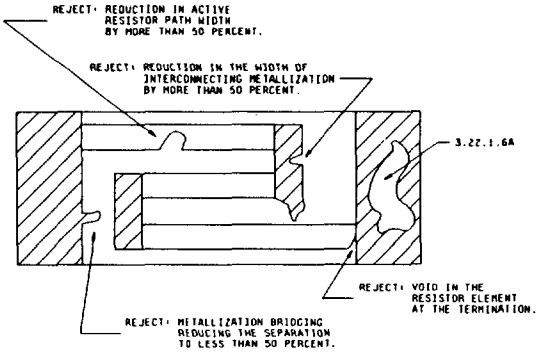
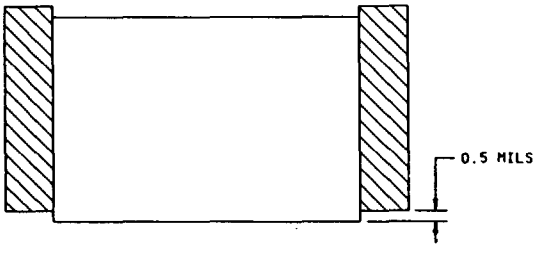


FIGURE 3. Metallization voids, bridging, missing metallization and resistor voids.

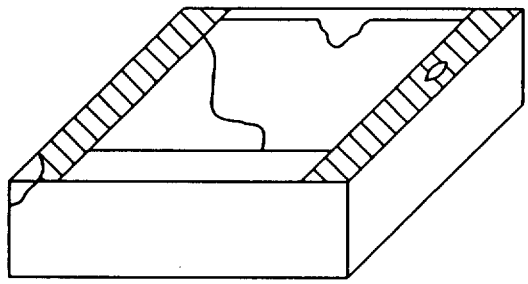


REJECT: MISALIGNMENT BETWEEN RESISTOR AND METALLIZATION GREATER THAN 0.5 MILS.

REJECT: MISALIGNMENT BETWEEN RESISTOR AND METALLIZATION GREATER THAN 0.5 MILS.

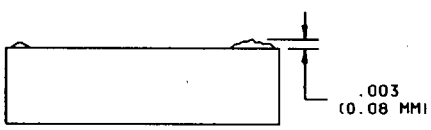
FIGURE 4. Metallization misalignment.

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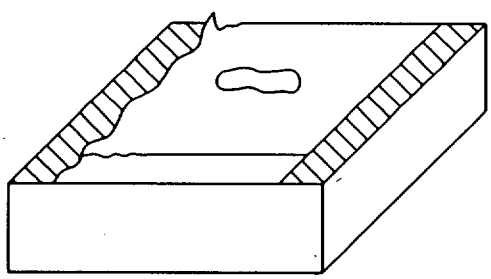
REJECT: ANY SCRATCHES ON RESISTOR AREA.

FIGURE 5. Resistor voids or scratches.



REJECT: EXCESSIVE BUILD UP OF TERMINATION ON ANY SURFACE MATERIAL GREATER THAN .003 INCH (0.08 MM) AND .008 INCH (0.20 MM) FOR SOLDERABLE AND BONDABLE LEADS, RESPECTIVELY.

FIGURE 6. Excessive metallization.



REJECT: ANY EVIDENCE OF PEELING, LIFTING OR BLISTERING.

FIGURE 7. Peeling, lifting, or blistering of film or metallization.

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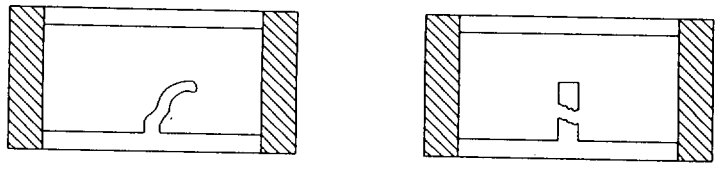
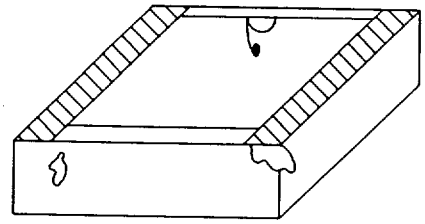
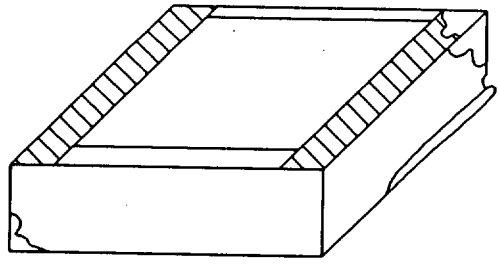


FIGURE 8. Laser trim faults.



REJECT: METALLIZATION CORROSION (LOCALIZED STAINS, CORROSION, DISCOLORIZATION, MOTTED METALLIZATION).

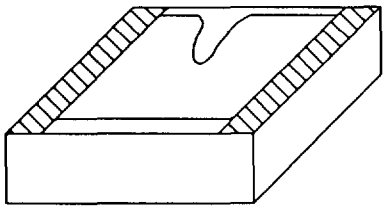
FIGURE 9. Metallization corrosion.



REJECT: MISSING BONDING PAD (ANY SHORTING BAR OR METAL PAD IS MISSING UNLESS BY DESIGN.

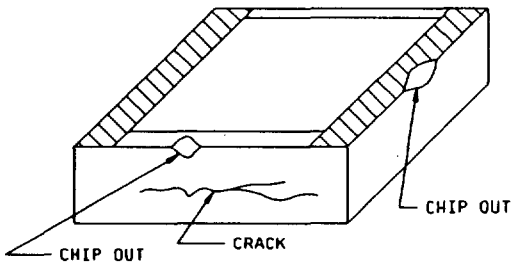
FIGURE 10. Missing metallization.

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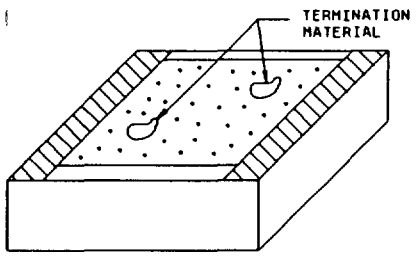
REJECT: VOID, NICK, OR BUBBLE OR CUT IN ACTIVE RESISTOR AREA GREATER THAN 50 PERCENT OF THE RESISTOR WIDTH EXCEPT FOR A TRIM CUT. THE USABLE RESISTANCE WIDTH FOR CHARACTERISTIC E, H, J, K AND M SHALL NOT BE LESS THAN 50 PERCENT OF THE ACTIVE RESISTOR ELEMENT.

FIGURE 11. Voids, nicks, bubble or cuts.



REJECT: ANY CRACK, CHIP OUT OR VOID IN THE SUBSTRATE GREATER THAN .003 INCH (0.08 MM) IN ANY ONE DIRECTION.

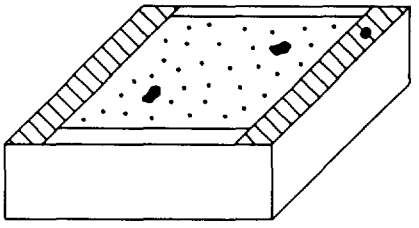
FIGURE 12. Cracks, chip outs, or voids.



REJECT: VISIBLE IMBEDDED FOREIGN PARTICLES IN ETHER THE RESISTOR FILM OR TERMINATION AREA. FOREIGN PARTICLES ARE THOSE THAT ARE NOT NORMAL TO RESISTOR DESIGN OR PROCESS.

FIGURE 13. Termination material.

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REJECT: VISIBLE IMBEDDED FOREIGN PARTICLES IN ETHER THE RESISTOR FILM OR TERMINATION AREA. FOREIGH PARTICLES ARE THOSE THAT ARE NOT NORMAL TO RESISTOR DESIGN OR PROCESS.

FIGURE 14. Imbedded particles.

3.23.4 Supplying to higher failure rate levels. A manufacturer may supply to all higher failure rate levels than to which they are qualified. Parts qualified to lower failure rate levels, with procuring activity approval, are substitutable as long as the label reflects the original failure rate level ordered (see table VII). Parts that are physically marked shall not be remarked unless specified in the contract or purchase order (see 6.2).

3.23.5 Supplying to lower resistance temperature characteristics and or lower resistances tolerances. Parts qualified to lower resistance temperature characteristics and/or lower resistance tolerance levels are substitutable, with procuring activity approval, for higher resistance temperature characteristic or higher tolerance levels as long as the label reflects the original resistance temperature characteristic and/or resistance tolerance (see table VII). Parts that are physically marked shall not be remarked unless specified in the contract or purchase order (see 6.2).

3.24 Workmanship. Resistors shall be processed in such a manner as to be uniform in quality and shall meet the requirements of 3.1, 3.4, 3.5, 3.5.1 through 3.5.3 inclusive, 3.23, as applicable, and be free from other defects that will affect life, serviceability, or appearance, and shall pass the visual inspection as specified on figures 3 through 14.

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TABLE VII. Substitution data.

Failure rate	Acceptable failure rate substitute
S (0.001)	
R (0.01)	S
P (0.1)	S, R
M (1.0)	S, R, P
Resistance tolerance	**Acceptable resistance tolerance substitute**
0.1	
0.5	0.1
1.0	0.1, 0.5
2.0	0.1, 0.5, 1.0
5.0	0.1, 0.5, 1.0, 2.0
10.0	0.1, 0.5, 1.0, 2.0, 5.0
Characteristic	**Acceptable characteristic substitute**
E	
H	E
K	
M	K

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations or tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Product assurance program. A reliability assurance program shall be established and maintained in accordance with MIL-STD-790. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.1.3 Statistical process control (SPC). A SPC program shall be maintained in accordance with EIA-557. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.2 Classification of inspection. The examination and testing of resistors shall be classified as follows:

a. Qualification inspection (see 4.4).

b. Verification of qualification (see 4.5).

c. Quality conformance inspection (see 4.6).

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4.3 Inspections conditions and precautions.

4.3.1 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the general requirements of MIL-STD-202.

4.3.2 Precautions. Adequate precautions shall be taken during inspection to prevent condensation of moisture on resistors, except during the moisture resistance temperature cycling tests.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3).

4.4.1 Sample. The number of sample units comprising a sample of resistors to be submitted for qualification inspection shall be as specified in the appendix to this specification. The sample shall be taken at random from a production run and shall be produced with equipment and procedures normally used in production, and which have been subjected to and passed the requirements of group A inspection (see 4.6.1.2). Qualification shall not be granted if group A inspection requirements are not complied with.

4.4.2 Test routine. Sample units shall be subjected to the qualification inspection specified in table VIII in the order shown. All sample units with the exception of ten sample units for group V inspection shall be subjected to the inspection of group I. The 182 sample units shall be divided as specified in table VIII for group II, group III, group IV, and group VI and subjected to the inspection for their particular group. Ten sample units shall be submitted to group VII, if applicable. Sample sizes and extent of qualification for characteristics shall be specified in the appendix to this specification.

As an option, the manufacturer may develop a control procedure for minimum stabilization times required for RTC testing. This procedure shall detail specific time elements required to reach thermal and electrical stability for specific part designs. Documentation of this procedure and supporting data shall be submitted to the qualifying activity for approval. Upon approval by the qualifying activity, the manufacturer may use this procedure for minimum stabilization times for RTC testing.

4.4.3 Defective. Defectives in excess of those allowed in table VIII shall be cause for refusal to grant qualification.

4.4.4 Failure rate level and quality level verification.

4.4.4.1 Procedure. FR qualification shall be in accordance with the general requirements of MIL-STD-690 and the following details:

a. Procedure I: Qualification at the initial FR level. Level M (1.0 percent) of FRSP-60 shall apply. Sample units shall be subjected to the qualification inspection specified in group IV, table VIII (see 4.4.2). Entire life test sample shall be continued on test to 10,000 hours as specified in 3.16.1 upon completion of the 2,000 hour qualification.

b. Procedure II: Extension of qualification to lower FR levels. To extend qualification to the R (0.01 percent) and S (0.001 percent) FR levels, data from two or more styles of similar construction may be combined.

c. Procedure III: Maintenance of FR level qualification. Maintenance period A of FRSP-10 shall apply. Regardless of the number of production lots produced during this period, the specified number of unit hours shall be accumulated to maintain qualification (see 4.6.2).

4.4.4.2 Quality level verification. The contractor is responsible for establishing a quality system to verify the PPM defect level of lots that are subjected to subgroup 3 tests of the group A inspections. The PPM defect level shall be based on a 6 month moving average. The contractor shall verify and report monthly the individual PPM categories (i.e., PPM-2 and PPM-3) and an overall PPM defect level (i.e., PPM-5) (see 3.3.2.2.1).

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4.5 Verification of qualification. Every 6 months, the manufacturer shall compile a summary of the results of quality conformance inspections in the form of a verification of qualification report, and forward it to the qualifying activity within 30 days after the end of the reporting period as the basis of continued qualification approval. In addition, the manufacturer shall immediately notify the qualifying activity whenever the FR data indicates that the manufacturer has failed to maintain the qualified FR level or the group C inspection data indicates failure of the qualified product to meet the requirements of this specification. Continuation of qualification shall be based on evidence that over the 6 month period the following has been met:

a. Verification by the qualifying activity that the manufacturer meets the requirements of MIL-STD-790.

b. The manufacturer has not modified the design of the item.

c. The specification requirements for the item have not been amended so as to affect the character of the item.

d. Lot rejection for group A, subgroup 2 and subgroup 5, inspection does not exceed 5 percent or one lot, whichever is greater.

e. Lot rejection for group B inspection does not exceed 5 percent or one lot, whichever is greater. (Subgroup 2, group B, is not included in the reject rate.)

f. The requirements of group C inspection are met.

g. The records of all failure rate tests combined substantiate that the M (1.0 percent), or P (0.1 percent) FR levels have been maintained or that the manufacturer continues to meet the R (0.01 percent) or S (0.001 percent) failure rate level for which qualified, although the total component hours of testing does not, as yet, meet the requirements of 4.4.4.1c.

h. The manufacturer shall provide documentation to the qualifying activity pertaining to PPM calculations including numbers of part types tested, individual PPM defect categories (i.e., PPM-2 and PPM-3) and the overall PPM defect rate (PPM-5). This information shall be submitted on a detail specification basis and based on a 6-month moving average. The manufacturer may group styles with qualifying activity approval.

If group C requirements were not met and the manufacturer has taken corrective action satisfactory to the Government, the forwarding of the verification of qualification report may be delayed until within 30 days after completion of retesting of the periodic quality conformance tests. In this case, the qualifying activity shall be notified of this condition within the time that the original verification of qualification report was due. All reports shall be certified by the responsible company official and the government inspector. The qualifying activity shall be contacted for the report format.

If group C test requires a comparison of "post test" readings with initial readings (delta measurements) the verification of qualification summary shall include the maximum and minimum delta changes for each inspection lot. For life testing delta R readings shall be reported at each interval in which readings are taken.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A and group B inspections. Except as specified in 4.6.2.5, delivery of products which have passed the group A and group B inspections shall not be delayed pending the results of the group C inspection. Group B inspection for preparation of delivery is not required when the qualifying activity has allowed group B testing to be performed annually (see table II).

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TABLE VIII. Qualification inspection.

Inspection	Requirement paragraph	Method paragraph	Number of samples	Number of defects allowed
Group I				
Visual and mechanical inspection	3.1, 3.4, 3.5, 3.5.1 through 3.5.3 inclusive, 3.22, and 3.23	4.7.1	All sample units except group V	0
DC resistance	3.8	4.7.2		
Group II				
Resistance to bonding exposure	3.13	4.7.7	10 highest 10 critical 10 lowest (30)	1
Resistance temperature characteristic	3.15	4.7.9		
Low temperature operation	3.10	4.7.4		
Short time overload	3.11	4.7.5		
High temperature exposure	3.12	4.7.6		
Group III				
Resistance to bonding exposure	3.13	4.7.7	10 highest 10 critical 10 lowest (30)	1
Moisture resistance	3.14	4.7.8		
Group IV				
Life	3.16	4.7.10.1	34 highest 34 critical 34 lowest (102)	1
Group V				
Solderability 1/	3.17	4.7.11	10 any value	1
Group VI				
Solderable mounting integrity 1/	3.18.1	4.7.12.1	10 any value	1
Bondable mounting integrity 2/	3.18.2	4.7.12.2	10 any value	
Wire bonding integrity 3/	3.18.3	4.7.12.3	10 any value (30)	
Group VII				
Resistance to solvents 4/	3.19	4.7.13	10 any value	0
Laser marking legibility 5/	3.20	4.7.14	10 any value (20)	

1/ Applicable to terminations B, R, and S.
2/ Applicable to terminations C, D, G, T and U.
3/ Applicable to terminations W and P.
4/ Test applicable for screen printed or color dot parts.
5/ Test applicable for laser marked parts.

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4.6.1.1 Inspection and production lot.

4.6.1.1.1 Inspection lot. An inspection lot shall consist of resistors of the same style, characteristic, and termination type and manufactured under essentially the same process and conditions with the same basic materials and offered for inspection at one time.

4.6.1.1.2 Production lot. A production lot shall consist of all resistors of the same style, nominal resistance value, resistance tolerance, resistance temperature characteristic, and termination material. Manufacture of all parts in the lot shall have been started, processed, assembled, and tested as a group. Lot identity shall be maintained throughout the manufacturing cycle.

4.6.1.1.3 Optional production lot control (Characteristics E and K only). A manufacturer that has a single design that meets all the requirements of both characteristics E and K may, with qualifying activity approval, use a different control system. All products shall be processed as E characteristic but may be subsequently supplied as K characteristic provided the manufacturer marks the unit packages accordingly and performs all other QCI tests for characteristic K.

4.6.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table IX and shall be made on the same set of sample units, in the order shown.

4.6.1.2.1 Subgroup 1 test (precap visual). The subgroup 1 test shall be performed on a production lot basis. Thirteen samples shall be subjected to subgroup 1. In the event of one or more failures, the lot shall be rejected.

4.6.1.2.1.1 Manufacturer's production inspection. If the manufacturer performs tests similar to those specified in subgroup 2, table IX, as the final step of his production process, group A, subgroup 2 inspection may be waived and the data resulting from the manufacturer's production tests may be used instead. Authority to waive the subgroup 2 inspection shall be granted by the qualifying activity only. The following criteria must be complied with:

a. Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than those specified for subgroup 2. Test conditions shall be equal to or more stringent than those specified for subgroup 2 tests.

b. Manufacturer subjects 100 percent of the product supplied under this specification to his production tests.

c. The parameters measured and the failure criteria shall be the same or more stringent than those specified herein.

d. The lot rejection criteria are the same or more stringent than those specified herein.

e. The manufacturer shall make available all information concerning the test procedures and instrumentation used in his production tests. The manufacturer shall also make available to the Government all records of all detail test data resulting from production tests.

f. Once approved, the manufacturer shall not change the test procedures or criteria without prior notification to and concurrence from the qualifying activity.

4.6.1.2.1.2 Rejected lots. In the event of one or more defects, and the production lot is rejected, the rejected production lot(s) shall be submitted to a 100 percent precap visual inspection as specified in 3.22.2. Resistors that pass precap visual inspection are available to continue group A testing.

4.6.1.2.2 Subgroup 2 tests. Subgroup 2 tests shall be performed on a production lot basis on 100 percent of the product supplied under this specification. Resistors that are out of resistance tolerance, or which experience a change in resistance greater than that permitted for the tests of this subgroup shall be removed from the lot. Lots having more than 5 percent total rejects, due to exceeding the specified resistance tolerance change limit shall not be furnished on contracts.

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TABLE IX. Group A inspection.

Inspection	Requirement paragraph	Method paragraph	Sampling procedure
Subgroup 1			
Precap visual inspection 1/	3.22	4.7.1	4.6.1.2.1
Subgroup 2			
Thermal shock	3.9	4.7.3	4.6.1.2.2
DC resistance	3.8	4.7.2	
Subgroup 3 (PPM)			
DC resistance (PPM-2)	3.8	4.7.2	4.6.1.2.3
Mechanical inspection (PPM-3)	3.1	4.7.1	
Subgroup 4			
Visual inspection 2/	3.1, 3.4, 3.5 through 3.5.3 inclusive, and 3.22	4.7.1	4.6.1.2.4
Subgroup 5			
Solderability	3.17	4.7.11	4.6.1.2.5
Subgroup 6			
Resistance to solvents 3/	3.19	4.7.13	4.6.1.2.6
Marking legibility 4/	3.20	4.7.14	

1/ Precap visual inspection shall be performed when the resistance element can not be inspected after passivation/glassivation in subgroup 4.
2/ Reference 3.22 is not applicable if precap visual inspection is completed in subgroup 1.
3/ Test applicable for color dot or screen printed parts.
4/ Test applicable for laser marked parts.

4.6.1.2.3 Subgroup 3 tests (PPM categories).

4.6.1.2.3.1 Sampling plan. Subgroup 3 tests shall be performed on an inspection lot basis. Samples subjected to subgroup 3 shall be selected in accordance with table X based on the size of the inspection lot, (NOTE: Larger samples may be inspected by the contractor in order to calculate PPM, however rejection of the lot shall be based on one or more defects). In the event of one or more failures, the lot shall be rejected. Equipment used to perform the subgroup 3 tests shall not be the same as those used in the subgroup 2, 100 percent inspection tests.

4.6.1.2.3.2 Rejected lots. The rejected lot shall be segregated from new lots that have passed inspection. The rejected lots shall be 100 percent inspected for those quality characteristics found defective in the sample and defects found shall be removed from the lot. A new sample of parts shall be randomly selected using the original sample size (see 4.6.1.2.3.1). If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.2.3.3 PPM calculations. PPM calculations shall be based on the results of the first sample check as prescribed in 4.6.1.2.3.1. Calculations and data exclusion shall be in accordance with EIA-554. (NOTE: PPM calculations shall not use data on the second sample submission).

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4.6.1.2.4 Subgroup 4 (visual inspection). The subgroup 4 test shall be performed on an inspection lot basis. A sample of parts shall be randomly selected in accordance with table X. If one or more defects are found, the lot shall be reworked or screened and defectives removed. After reworking or screening and removal of defectives, a new sample of parts shall be randomly selected in accordance with table X. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.2.5 Subgroup 5 (solderability).

4.6.1.2.5.1 Samples. Thirteen samples shall be selected randomly from each inspection lot and subjected to the subgroup 5 solderability test. The manufacturer may use electrical rejects from the subgroup 2 screening tests for all or part of the samples to be used for the solderability testing. If there are one or more defects, the lot shall be considered to have failed.

4.6.1.2.5.2 Rejected lots. In the event of one or more defects, the inspection lot is rejected. Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 3.17. An additional 13 samples shall be randomly selected from each production lot to be tested. If there are one or more defects in the production lot sample, that production lot shall be considered to have failed and that product will not be supplied to the requirements of this specification. Individual production lot(s) that pass the solderability test are available for shipment.

TABLE X. Sampling plans for subgroup 3 and subgroup 4.

Lot size	Sample size for subgroup 3	Sample size for subgroup 4
1 to 13	100 percent	100 percent
14 to 125	100 percent	13
126 to 150	125	13
151 to 280	125	20
281 to 500	125	29
501 to 1,200	125	34
1,201 to 3,200	125	42
3,201 to 10,000	192	50
10,001 to 35,000	294	60
35,001 to 150,000	294	74
150,001 to 500,000	345	90
500,001 and over	435	102

4.6.1.2.5.3 Disposition of samples. The solderability test is considered a destructive test and samples submitted to the solderability test shall not be supplied on the contract.

4.6.1.2.6 Subgroup 6. Eight samples shall be selected randomly from each inspection lot and subjected to the subgroup 6 tests. The manufacturer may use electrical rejects from the subgroup 2 screening tests for all or part of the samples to be used for the subgroup 6 testing. If there are one or more defects, the lot shall be considered to have failed.

4.6.1.2.6.1 Rejected lots. In the event of one or more defects, the inspection lot is rejected. Each production lot that was used to form the failed inspection lot shall be individually submitted to the subgroup 6 test as required in 3.19 or 3.20. An additional eight samples shall be randomly selected from each production lot and shall be tested. If there are one or more defects in the production lot sample, that production lot shall be considered to have failed and that product will not be supplied to the requirements of this specification. Individual production lots that pass the minimum marking test are available for shipment.

4.6.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table XI in the order shown, and shall be performed on samples from lots which have been subjected to and have passed the group A inspection.

4.6.1.3.1 Sampling plan.

4.6.1.3.1.1 Subgroup 1. Twenty sample units of parts shall be randomly selected from the inspection lot. If one or more defects are found, the lot shall be reworked or screened and defectives removed. After reworking or screening and removal of defectives, a new sample of 20 parts shall be randomly selected. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

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TABLE XI. Group B inspection.

Inspection	Requirement paragraph	Method paragraph	Number of samples
Subgroup 1 1/			
Resistance temperature characteristic	3.15	4.7.9	20 samples
Short time overload	3.11	4.7.5	
Subgroup 2 2/			
Solder mounting integrity 3/	3.18.1	4.7.12.1	10 samples
Bondable mounting integrity 4/	3.18.2	4.7.12.2	10 samples
Wirebonding integrity 5/	3.18.3	4.7.12.3	10 samples

1/ If the manufacturer can demonstrate that these tests have been performed five consecutive times with zero failures, the frequency of these tests, with the approval of the qualifying activity can be performed on a annual basis. If the design, material, construction, or processing of the part is changed or, if there are any quality problems or failures, the qualifying activity may require resumption of the original test frequency.

2/ If the manufacturer can demonstrate that this test has been performed five consecutive times with zero failures, this test, with the approval of the qualifying activity, can be deleted. The manufacturer, however, shall perform this test every three years after the deletion as part of long term design verification. If the design, material, construction, or processing of the part is changed or, if there are any quality problems, the qualifying activity may require resumption of the specified testing. Deletion of testing does not relieve the manufacturer from meeting the test requirement in case of dispute.

3/ Applicable to terminations B, R, and S.

4/ Applicable to terminations C, D, G, T, and U.

5/ Applicable to terminations W and P.

4.6.1.3.1.2 Subgroup 2. A sample of ten parts shall be randomly selected from the inspection lot. If one or more defects are found, the lot shall be reworked or screened and defectives removed. After reworking or screening and removal of defectives, a new sample of ten parts shall be randomly selected. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.3.2 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or purchase order.

4.6.2 Group C inspection. Group C inspection shall consist of the tests specified in table XII, in the order shown. Group C inspection shall be made on sample units selected from lots which have passed group A and group B inspection. Group C samples shall be representative of production. The allowable number of defectives shall be as indicated in table XII. The assigned periodic tests shall be initiated within 30 days of the specified periodic inspection.

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TABLE XII. Group C inspection.

Inspection	Requirement paragraph	Method paragraph	Number of samples	Number of defects allowed
Monthly Subgroup 1				
Life (ER, ceramic substrate)	3.16	4.7.10.1	See 4.6.2.1.1	See 4.6.2.1.1
Subgroup 2 1/				
Thermal shock Low temperature operation	3.9 3.10	4.7.3 4.7.4	30	1
Quarterly Subgroup 1				
Resistance to bonding exposure Moisture resistance	3.13 3.14	4.7.7 4.7.8	30	1
Subgroup 2 1/				
High temperature exposure	3.12	4.7.6	30	1
Subgroup 3				
Life (FR-4 board)	3.16	4.7.10.2	15 (5 high, 5 critical, 5 low)	1

1/ If the manufacturer can demonstrate that these tests have been performed five consecutive times with zero failures, the frequency of these tests, with the approval of the qualifying activity can be performed on a annual basis. If the design, material, construction, or processing of the part is changed or, if there are any quality problems or failures, the qualifying activity may require resumption of the original test frequency.

4.6.2.1 Monthly.

4.6.2.1.1 Subgroup 1. Samples shall be accumulated from inspection lots and placed on extended life test in accordance with 4.7.10 once a month for the full 10,000 hour life test. A sufficient number of samples shall be selected by the manufacturer so that the maintenance of failure rate requirements comply with the specified maintenance period. Samples shall be representative of production of individual styles, characteristics, and terminations in the maintenance period. The manufacturer has the option, with qualifying activity approval of selecting samples from RTC groups versus individual characteristics (Group I: E or H; Group II: K or M). As far as practical, the manufacturer shall select the resistance values so that all resistance decades produced during the maintenance period are represented. The accumulated data shall be used for maintenance or extension of failure rate qualification.

4.6.2.1.2 Subgroup 2. Subgroup 2 tests shall be performed on a sample size of 30 units of each style, characteristic, and termination with one defective allowed for each group of 30.

4.6.2.2 Quarterly.

4.6.2.2.1 Subgroup 1. Every 3 months, 30 units of each style, characteristic, and termination material (any resistance value) shall be subjected to quarterly tests specified in table XII, in the order shown.

4.6.2.2.2 Subgroup 2. Every 3 months, 30 units of each style, characteristic, and termination material (any resistance value) shall be subjected to quarterly tests specified in table XII, in the order shown.

4.6.2.2.3 Subgroup 3. Every 3 months, 15 units of each style, characteristic, and termination material, except W and P, shall be subjected to the subgroup 3 quarterly tests as specified in table XII. As an option, to performing testing of each characteristic and termination finish, the manufacturer may construct groups based on the guidance in table XVII (NOTE: Combination of styles is not permitted).

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4.6.2.3 Noncompliance. If a sample fails to pass group C inspection, the contractor shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Group A and group B inspections may be reinstituted; however, final acceptance shall be withheld until group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.6.3 Inspection of packaging. Sample packages and packs and the inspection of the preservation packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of MIL-R-39032.

4.7 Methods of inspection.

4.7.1 Visual and mechanical inspection, and chip mounting.

4.7.1.1 Visual and mechanical inspection. Unless otherwise specified resistors shall be examined under 30X to 60X magnification, but in case of conflict 30X will be the referee power, unless otherwise specified, to verify that the materials, design, construction, physical dimensions, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4, 3.5, 3.5.1 through 3.5.3 inclusive, 3.22 to 3.22.5d inclusive and 3.23).

4.7.1.2 Chip mounting procedures. When specified herein, the chip resistor shall be mounted on a test board as described in 4.7.1.3. For those test procedures where mounting requirements are unspecified, the chip resistors may be tested unmounted using pressure contacts.

4.7.1.3 Specified mounting. When specified in the test procedure, the chip resistor shall be mounted on a suitable test board as specified on figure 15. The test board material shall be such that it shall not be the cause of, nor contribute to any failure of a chip resistor in any of the tests for which it may be used. The test board shall be prepared with metallized surface land areas of proper spacing so that a test board surface area of at least four times the resistor chip surface area is provided for each resistor chip mounted. The metallized surface land areas shall be designed in a pattern to accommodate a number of resistor chips. The metallization material shall be compatible with the bonding technique to be employed and the material used on the chip termination. The method of chip mounting for the different termination materials shall be as follows:

a. All terminations (except P and W). All chip resistors (except P and W) termination materials shall be mounted on the test board by soldering the chip terminations directly to the test board metallized land areas in the following manner:

(1) Solder paste and flux shall be applied to the terminations of each chip of code letters B, C, D, R, S, T, and U (see 1.2.4).

(2) All chips shall be placed across the metallized land areas of the test board with contact between the chip terminations and board land areas only. The use of adhesive to keep chips in place during mounting operation is allowable.

(3) For resistance to bonding exposure, the test board with all chips in position shall be placed on a suitable heat transfer unit (molten solder, hot plate, tunnel oven, etc.) with the temperature maintained at 260°C ±10°C for 5 seconds ±2 seconds after reflow.

(4) For load life, the test board with all chips in position shall be placed on a suitable heat transfer unit (molten solder, hot plate, tunnel oven, etc.) with suitable time and temperature to assure proper reflow mounting.

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b. Termination material "W" and "P" (wire bondable). Chip resistors with wire bondable terminations shall be mechanically attached to the test board and electrically connected by wire bonding flying lead interconnections between the chip termination and the test board metallized land areas. The interconnecting lead shall be .001 inch (0.03 mm) in diameter gold wire. The chip shall be mechanically or adhesively mounted by any procedure which will not be the cause of or contribute to any failures of the chip resistor in any test.

c. Alternative mounting for terminations D, C, T, and U. As an alternative to 4.7.1.3a(1), chip resistors be mechanically attached to the test board with type 2 epoxy in accordance MMM-A-1931. The epoxy shall be cured at a minimum temperature of 150°C. The minimum curing time shall be 30 minutes.

4.7.2 DC resistance (see 3.8). The dc resistance shall be measured in accordance with method 303 of MIL-STD-202. The following details and exceptions shall apply:

a. Measuring apparatus: Measuring instruments used for initial and final readings of this test, can be of different styles or models provided the performance is equivalent. All test equipment shall be calibrated in accordance with MIL-STD-45662.

b. Test voltage for bridges: Measurements of resistance shall be made by using the test voltages up to the maximum values specified in table XIII. The test voltage chosen, whether it be the maximum or a lower voltage which would still provide the sensitivity required, shall be applied across the terminals of the resistor. This same voltage shall be used whenever a subsequent resistance measurement was made.

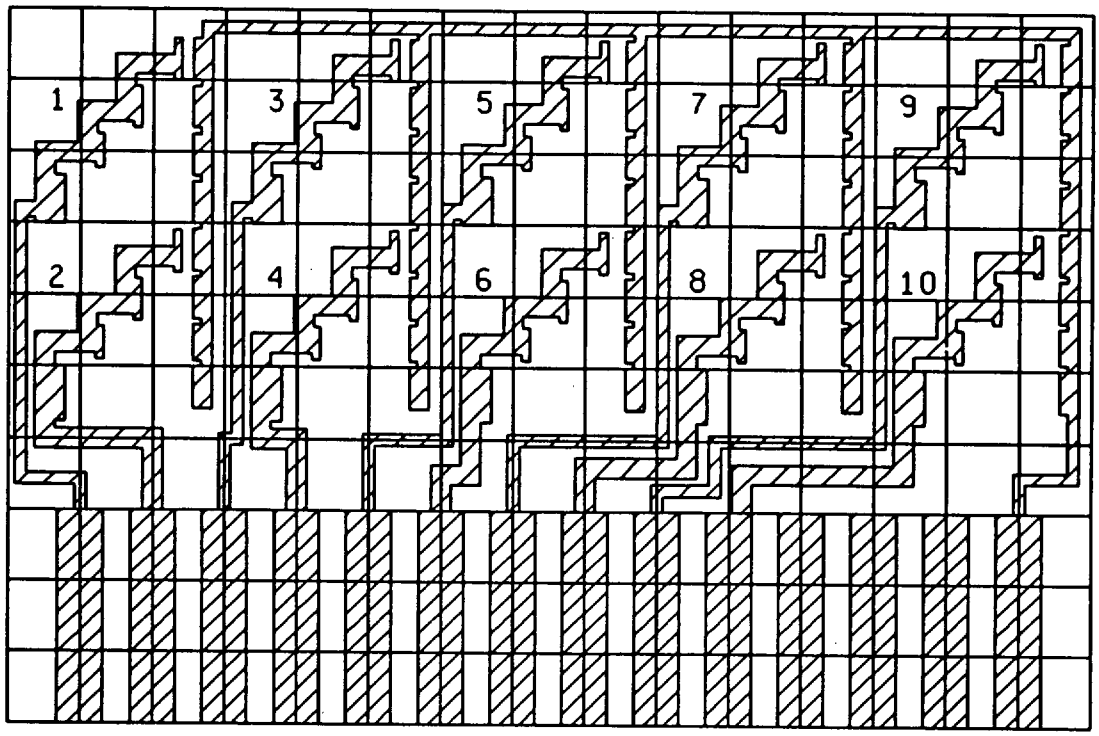
c. Measurement energy for electronic test equipment: The measurement energy applied to the unit under test shall not exceed 10 percent of the 25°C rated wattage times 1 second.

d. Temperature: The dc resistance test specified in group I of table VIII shall be performed at 25°C ±2°C. For all other tests, unless otherwise specified herein, the temperature at which subsequent and final resistance measurements are made in each test shall be within ±2°C of the temperature at which the initial resistance measurement was made.

TABLE XIII. Standard dc resistance test voltages.

Nominal resistance	Maximum test voltage			
	10 to 24 milliwatts	25 to 99 milliwatts	100 to 225 milliwatts	226 to 1,000 milliwatts
ohms	Volts	Volts	Volts	Volts
5.60 to 9.88	0.2	0.3	0.7	1.0
10 to 98.8	0.3	0.3	1.0	1.0
100 to 988	1.0	1.0	1.0	3.0
1,000 to 9,880	3.0	3.0	3.0	10.0
10,000 to 98,880	10.0	10.0	10.0	30.0
100,000 or higher	30.0	30.0	30.0	100.0

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NOTES:
1. Suggested ceramic test board: 1.000 inch (25.4 mm) x 1.500 inch (38.10 mm) x 0.025 inch (0.64 mm) thick.
2. Suggested fiberglass base (FR-4) test board: 1.000 inch (25.4 mm) x 1.500 inch (38.10 mm) x .0625 inch thick (1.5875 mm).
3. The contacts fit a printed circuit board connector with .100 inch (2.54 mm) in contact center.
4. Short lead wire may be soldered on and it will then fit a microcircuit connector, or test lead wires may soldered on for direct measurement.
5. Small grid square are .010 inch (0.25 mm).

FIGURE 15. Suggested test board.

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4.7.3 Thermal shock (see 3.9). Resistors shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions apply:

a. Mounting: Resistors shall not be mounted. Resistors may be placed in metal baskets, vials, or other apparatus as long as resistors are subjected to the specified temperature extremes. (For group C inspection only, as an option to the manufacturers, resistors may be mounted or unmounted.)

b. Measurement before cycling:

(1) Qualification and group A tests: DC resistance shall be measured as specified in 3.8, except at 25°C ±5°C. The dc resistance shall be within the specified resistance tolerance of the nominal resistance value.

(2) Group C test: DC resistance shall be measured as specified in 3.8.

c. Test condition F (except temperatures shall be 150°C +10°C, -0°C and -65°C +0°C, -10°C).

d. Measurement after cycling:

(1) Qualification and group A tests: After stabilization at room temperature, the dc resistance shall again be measured as specified in 3.8, except at 25°C ±5°C.

(2) Group C test: After stabilization at room temperature, the dc resistance shall again be measured as specified in 3.8. Following the test, the resistors shall be examined for evidence of mechanical damage.

4.7.4 Low temperature operation (see 3.10). A dc resistance measurement shall be made as specified in 3.8, mounted on a ceramic test board as specified in 4.7.1.3. The resistors shall be placed in a cold chamber maintained at -65°C +0°C, -5°C. After one hour of stabilization at this temperature, full rated continuous working voltage as specified in 3.7 shall be applied for 45 minutes +5 minutes, -0 minutes. Fifteen minutes +5 minutes, -0 minutes after the removal of the voltage, the resistors shall be removed from the chamber and maintained at a temperature of 25°C ±5°C until stabilized; the dc resistance shall be measured as specified in 3.8. Resistors shall be examined for evidence of mechanical damage.

4.7.5 Short time overload (see 3.11). The dc resistance shall be measured as specified in 3.8, and mounted on a ceramic test board as specified in 4.7.1.3. A dc test potential, 2.5 times the rated continuous working voltage but not exceeding twice the maximum voltage (see 3.1) shall be applied for 5 seconds to the resistor under the following conditions:

a. Resistors on test board shall be mounted so that the larger space of the resistor is on a horizontal plane.

b. In still air, with no circulation other than that created by the heat of the resistors being operated.

Thirty minutes +15 minutes, -0 minutes after removal of the test potential, the dc resistance shall again be measured as specified in 3.8. Resistors shall be examined for evidence of arcing, burning, and charring.

4.7.6 High temperature exposure (see 3.12). Resistors shall be mounted on a ceramic test board as specified in 4.7.1.3 and the dc resistance shall be measured as specified in 4.7.2. The resistors shall then be subjected to a continuous temperature of 150°C ±5°C for a period of 100 hours ±4 hours, in a test chamber with forced air circulation. The resistors shall be stabilized at 25°C ±3°C after which the dc resistance shall be measured. The change in resistance shall not exceed the value specified in 3.12.

4.7.7 Resistance to bonding exposure (see 3.13). The dc resistance of the chip resistor shall be measured as specified in 3.8. The chip resistor shall be mounted on a ceramic test board in accordance with 4.7.1.3. The test board with resistors mounted shall be stabilized at room temperature after which the dc resistance shall again be measured as specified in 3.8. The change in resistance shall not exceed the value specified in 3.13.

4.7.8 Moisture resistance (see 3.14). Resistors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Chip resistor sample units shall be mounted on a ceramic test board as specified in 4.7.1.3.

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b. Initial measurement: Immediately following the initial conditioning period, dc resistance of each chip on the test board shall be measured as specified in 3.8.

c. Polarization and loading voltage: No voltage shall be applied.

d. Subcycle: Step 7b shall not be applicable. Step 7a shall be performed during any five of the first nine cycles only.

e. Measurements at high humidity: None.

f. Final measurements: Upon completion of step 6 of the final cycle, the resistors shall be removed from the chamber and air dried for 30 minutes ±15 minutes. The dc resistance shall then be measured in accordance with 3.8.

g. Examination after test: Resistors shall be examined for evidence of electrical and mechanical damage.

4.7.9 Resistance temperature characteristic (see 3.15). Resistors shall be tested in accordance with method 304 of MIL-STD-202. The following details and exceptions shall apply:

a. Reference temperature: Room ambient temperature.

b. Test temperature: In accordance with table XIV.

c. Accuracy of temperature measurement: Resistors shall be maintained for 30 minutes to 45 minutes within 3°C at each of the test temperatures listed in table XIV. This tolerance shall be maintained on the established test temperatures.

d. Mounting: Resistors shall be mounted on a ceramic test board as specified in 4.7.1.3.

TABLE XIV. Resistance temperature characteristic.

Sequence	Temperature
	Group B acceptance inspection 1/
	°C
1	2/ 25 ±3
2	-55 ±3
3	2/ 25 ±3
4	125 ±3

1/ At the option of the manufacturer, the reverse sequence may be as follows:
1 - 1/ 25 ±3
2 - 125 ±3
3 - 1/ 25 ±3
4 - -55 ±3

2/ This temperature shall be considered the reference temperature for each of the succeeding temperatures.

As an option, the manufacturer may develop a control procedure for minimum stabilization times required for RTC testing. This procedure shall detail specific time elements required to reach thermal and electrical stability for specific part designs. Documentation of this procedure and supporting data shall be submitted to the qualifying activity for approval. Upon approval by the qualifying activity, the manufacturer may use this procedure for minimum stabilization times for RTC testing.

4.7.10 Life (see 3.16).

4.7.10.1 Life test (ceramic boards). Resistors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

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a. Method of mounting: Chip resistors sample units shall be mounted on a test board as specified in 4.7.1.3.

b. Test temperature and tolerance: 70°C ±5°C.

c. Initial resistance measurement of mounted resistors: Measurements may be made inside or outside the chamber.

(1) Inside chamber: When measurements are to be made inside the chamber, the initial dc resistance shall be measured after mounting at the applicable test temperature, after temperature stabilization, and within 8 hours of exposure of the resistors to the test temperature. This initial measurement shall be used as the reference dc resistance for all subsequent measurements under the same condition.

(2) Outside chamber: When measurements are to be made outside the chamber, the initial dc resistance shall be measured after mounting at room temperature. This initial measurement shall be used as the reference dc resistance for all subsequent measurements under the same condition.

d. Operating conditions: Rated dc continuous working voltage or filtered or nonfiltered full wave rectified ac voltage shall be applied intermittently, 90 minutes on and 30 minutes off, for the applicable number of hours (see 4.7.10f) and at the applicable test temperature. "On time" shall be three quarters of the total elapsed time. During the "on" cycle, the voltage shall be regulated and controlled to maintain ±5 percent of the rated continuous working voltage.

e. Test condition: Two thousand hours elapsed time for qualification inspection with all samples continued to 10,000 hours. Ten thousand hours for failure rate level inspection group C.

f. Measurements during test:

(1) Qualification inspection: DC resistance shall be measured at the end of the 30 minutes off periods after 250 hours +72 hours, -24 hours; 500 hours +72 hours, -24 hours; 1,000 hours +72 hours, -24 hours; and 2,000 hours +96 hours, -0 hours have elapsed.

(2) Extended life testing: DC resistance shall be measured at the end of the 30 minutes off periods after 250 hours +72 hours, -24 hours; 500 hours +72 hours, -24 hours, 1,000 hours +72 hours, -24 hours; and 2,000 hours +96 hours, -48 hours, and every 2,000 hours +96 hours, -48 hours thereafter until the required 10,000 hours have elapsed. Measurements shall be made as near as possible to the specified time but may be adjusted so that measurements need not be made during other than normal workdays.

(3) Measurements outside of the chamber: When measurements are made outside the chamber, resistors shall be outside of the chamber, for a minimum of 45 minutes and stabilized before measurement.

g. Examination after test: Resistors shall be examined for evidence of mechanical damage.

4.7.10.2 Life test (fiberglass base (FR-4) boards). Resistors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Method of mounting: Chip resistors sample units shall be mounted on a test board as specified in 4.7.1.3.

b. Test temperature and tolerance: 70°C ±5°C.

c. Initial resistance measurement of mounted resistors: Measurements may be made inside or outside the chamber.

(1) Inside chamber: When measurements are to be made inside the chamber, the initial dc resistance shall be measured after mounting at the applicable test temperature, after temperature stabilization, and within 8 hours of exposure of the resistors to the test temperature. This initial measurement shall be used as the reference dc resistance for all subsequent measurements under the same condition.

(2) Outside chamber: When measurements are to be made outside the chamber, the initial dc resistance shall be measured after mounting at room temperature. This initial measurement shall be used as the reference dc resistance for all subsequent measurements under the same condition.

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d. Operating conditions: Rated dc continuous working voltage or filtered or nonfiltered full wave rectified ac voltage shall be applied intermittently, 90 minutes on and 30 minutes off, for the applicable number of hours (see 4.7.10.1f) and at the applicable test temperature. "On time" shall be three quarters of the total elapsed time. During the "on" cycle, the voltage shall be regulated and controlled to maintain ±5 percent of the rated continuous working voltage.

e. Test condition: Two thousand hours elapsed time for qualification inspection with all samples continued to 10,000 hours. Ten thousand hours for failure rate level inspection group C.

f. Measurements during test:

(1) Qualification inspection: DC resistance shall be measured at the end of the 30 minutes off periods after 250 hours +72 hours, -24 hours; 500 hours +72 hours, -24 hours; 1,000 hours +72 hours, -24 hours; and 2,000 hours +96 hours, -0 hours have elapsed.

(2) Extended life testing: DC resistance shall be measured at the end of the 30 minutes off periods after 250 hours +72 hours, -24 hours; 500 hours +72 hours, -24 hours, 1,000 hours +72 hours, -24 hours; and 2,000 hours +96 hours, -48 hours, and every 2,000 hours +96 hours, -48 hours thereafter until the required 10,000 hours have elapsed. Measurements shall be made as near as possible to the specified time but may be adjusted so that measurements need not be made during other than normal workdays.

(3) Measurements outside of the chamber: When measurements are made outside the chamber, resistors shall be outside of the chamber, for a minimum of 45 minutes and stabilized before measurement.

g. Examination after test: Resistors shall be examined for evidence of mechanical damage.

4.7.11 Solderability (applicable to B, R, and S terminations) (see 3.17). Resistors shall be tested in accordance with method 208 of MIL-STD-202. Both end terminations shall be immersed completely one at a time or both at the same time by dipping the entire chip in the solder pot. The following details and exceptions shall apply:

a. Apparatus:

(1) A solder pot capable of keeping the solder temperature to ±5°C of setpoint.

(2) A device to keep the specimen submerged in solder for the required time.

(3) Optical equipment capable of 30X magnification.

(4) A laboratory beaker or appropriate laboratory apparatus that is capable of:

(a) Supporting a nonmetallic device.

(b) Maintaining the test specimens at a distance of 1.5 inches to 2.5 inches from the surface of the boiling DI or distilled water while, at the same time exposing them to the full flow of steam generated.

b. The requirement for standard solderable wire application shall not apply.

c. The immersion and emersion rates shall not apply.

d. The specimen shall be totally submerged for at least 5 seconds.

4.7.12 Mounting integrity.

4.7.12.1 Solder mounting integrity. The resistor chips shall be prepared as specified in 4.7.1.3a. A force shall be applied to all solderable units. For styles RM0502, RM0505, RM1005, RM1505, RM0705, and RM1206, the force applied shall be 2 kilograms. For styles RM2208, RM2010, RM2512, and RM1010 the force applied shall be 3 kilograms. The force shall be applied to the edge of the chip as shown on figure 16 for a minimum 30 seconds. The resistor shall be examined for evidence of mechanical damage. The pusher width shall be a minimum of 30 percent of the length and the maximum width shall be 70 percent of the chip length.

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4.7.12.2 Bondable mounting integrity. The resistor chips shall be prepared as specified in 4.7.1.3c. A force shall be applied to all bondable units. For styles RM0502, RM0505, RM1005, RM1505, RM0705, and RM1206, the force applied shall be 1 kilogram. For styles RM2208, RM2010, RM2512, and RM1010, the force applied shall be 1.5 kilograms. The force shall be applied to the edge of the chip as shown on figure 16 for a minimum 30 seconds. The resistor shall be examined for evidence of mechanical damage. The pusher width shall be a minimum of 30 percent of the resistor length and the maximum width shall be 70 percent of the chip length.

4.7.12.3 Wirebonding integrity. The chip resistor shall be prepared as specified in 4.7.1.3b. A pull of 4 grams shall be applied by inserting a hook under the lead wire at approximately the center of the wire as shown in figure 17. The force shall be applied at a 90 degree angle to the surface of the chip, one lead at a time for 30 seconds each.

4.7.13 Resistance to solvents (see 3.19). Resistors shall be tested in accordance with method 215 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Unmounted.

b. The marked portion of the resistor shall be brushed.

c. The number of sample units shall be as specified in table IX.

d. Resistors shall be inspected for mechanical damage and legibility of minimum marking.

4.7.14 Laser marking legibility test (see 3.20). Resistors shall be coated with .005 inch (0.13 mm) minimum of silicon resin insulated compound, type SR of MIL-I-46058. After curing, coated resistors shall be examined for legibility under normal production room lighting by an inspector at 10X magnification.

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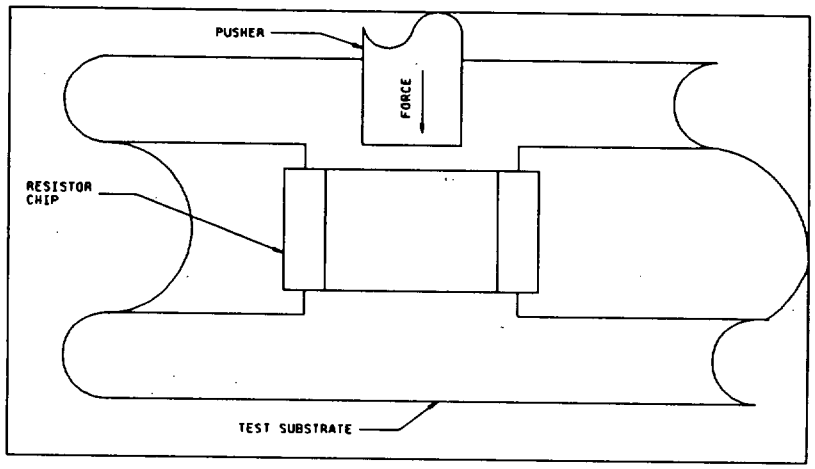


FIGURE 16. Mounting integrity test fixture.

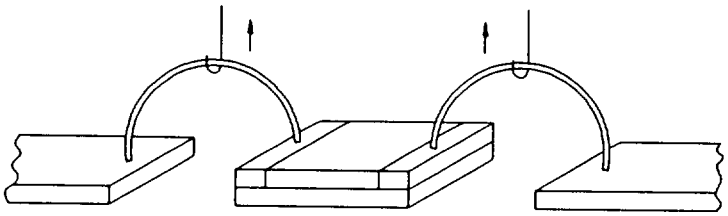


FIGURE 17. Weldable style units.

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5. PACKAGING

5.1 Packaging requirements. Resistors shall be prepared for delivery in accordance with MIL-R-39032.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Chip resistors are intended to be used in thin or thick film hybrid circuits where micro-circuitry is indicated and in surface mount applications.

6.2 Acquisition requirements. Acquisition requirements should specify the following:

a. Title, number, and date of this specification.

b. Title, number, and date of the applicable detail specification, and the complete PIN (see 1.2.1 and 3.1).

c. Termination configuration area (see 3.5.2).

d. Substrate bonding (see 3.5.3).

e. Minimum marking: if not specified, chip resistors will be supplied unmarked.

f. Allowable substitution (see 3.23.4 and 3.23.5).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award or contract, qualified for inclusion in the applicable QPL whether or not such products have been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the QPL is the US Army Research Laboratory, ATTN: AMSRL-PS-DC, Fort Monmouth, NJ 07703-5601; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-ELS), 1507 Wilmington Pike, Dayton, Ohio 45444-5764.

6.4 Application notes.

6.4.1 Mounting of resistors. Resistors may be mounted individually on a test board, fiber glass base, and connected to conductor areas by means of solder preforms, conductive cement, or wire bonding. Users are cautioned that the substrate and chip component must have compatible temperature coefficients of expansion to reduce the strain level imposed on the solder connections. They may also be directly connected to other components on the same test board by means of wire bonding, using the test board as a base or carrier for the resistor.

6.4.2 Stacking of resistors. Stacking is generally discouraged since experience indicates that failure may occur due to electrolytic action in the bonding adhesive. This action occurs from the increase in temperature caused by stacking in conjunction with the operating voltage across the resistor. The bonding adhesive at the resistor chip interface chemically reacts with the active resistor material changing the noise and resistance values and eventually the resistor will open. When stacking resistors, care should be taken to compensate for the lower heat dissipation capabilities by derating the wattage rating.

6.4.3 Close tolerance resistors. Close tolerance resistors (i.e., ±0.1 percent) should be mounted on a test board by a method which produces the least heating effect over a short time to avoid permanent changes in resistance.

6.4.4 Resistor film orientation. It is recommended that the film side of resistors be mounted up when they are placed onto printed circuit boards. This is due to the effect of thermal stressing on epoxies used to hold them in place when mounted on the bottom of printed circuit boards. Resistor films can be torn from the substrate during thermal cycling, substantially altering the value of the resistor.

6.4.5 Tin plated finishes. Tin plating is prohibited (see 3.5.2.2) since it may result in tin whisker growth. Tin whisker growth could adversely affect the operation of electronic equipment systems. For additional information on this matter refer to ASTM B545.

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6.5 Noise level test. This specification does not include a noise test. If the user requires a noise test, it shall be performed in accordance with method 308 of MIL-STD-202.

6.6 Electrostatic charge effects. Under relatively low humidity conditions, some types of film resistors, particularly those with small dimensions and high sheet resistivity materials, are prone to sudden significant changes in resistance (usually reductions in value) and to changes in temperature coefficient of resistance as a result of discharge of static charges built up on associated objects during handling, packaging, or shipment. Substitution of more suitable implements and materials can help minimize this problem. For example, use of cotton gloves, static eliminator devices, air humidifiers, and operator and work bench grounding systems can reduce static buildup during handling. Means of alleviating static problems during shipment include elimination of loose packaging of resistors and use of metal foil and anti-static (partly conducting) plastic packaging materials.

6.7 Terminal substitution data. For existing design and logistical support of inactivated termination R (solderable pretinned wraparound terminations), termination B (solderable base metallization barrier metal, solder coated) shall be used as a substitute for termination R.

6.8 Subject term (key word) listing.

Chip resistors
Film resistors
Surface mount

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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APPENDIX

PROCEDURES FOR QUALIFICATION INSPECTION

10. SCOPE

10.1 Scope. This appendix details the procedure for submission of samples with related data, for qualification inspection of resistors covered by this specification. The procedure for extending qualification of the required sample to other resistors covered by this specification is also outlined herein. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this document.

30. SUBMISSION

30.1 Sample. A sample consisting of 222 sample units, 74 of the lowest, 74 of the highest, and 74 of the critical or nearest to critical value (see table XV), and in each style and characteristic, termination material, and resistance tolerance for which qualification is sought, shall be submitted and subjected to the inspection of table VIII. If no critical value is specified, the sample will be equally divided between the highest and lowest resistance values. Ten sample units of any resistance value or characteristic shall be submitted and subjected to the test of group V. Twenty sample units of any resistance value shall be submitted to tests of group VI, when applicable. Thirty sample units shall be submitted to the test in group VII. If qualification to a closer resistance tolerance than submitted above is desired, 30 additional sample units of the closer resistance tolerance shall be subjected to the following tests (see table XVI).

TABLE XV. Sample size for qualification inspection.

Inspection	Requirement paragraph	Method paragraph	Number of sample units for inspection	Number of defectives allowed
Visual and mechanical inspection	3.1, 3.4, 3.5, 3.5.1 through 3.5.3 inclusive, 3.22 through 3.23, inclusive	4.7.1	30 — 10 highest, 10 critical, 10 lowest	0
DC resistance	3.8	4.7.2		
Thermal shock	3.9	4.7.3		

30.1.1 Qualification to extended resistance ranges. For qualification to the extended resistance ranges of higher resistance tolerances, 74 additional samples of high or low resistance shall be subjected to the tests of group I, group II, group III, and group IV of table VIII.

30.2 Test data. When examinations and tests are to be performed at a government laboratory, prior to submission, all sample units shall be subjected to all of the tests indicated as nondestructive in table IX. Each submission shall be accompanied by test data obtained from these tests. The performance of the destructive examinations and tests by the manufacturer on a duplicate set of sample units is encouraged, although not required. All test data shall be submitted in duplicate.

30.3 Description of items. The manufacturer shall submit a detailed description of the resistors being submitted for inspection, including materials used for the resistance element and the protective enclosure or coating. After qualification has been granted, no changes shall be made in materials, design, or construction without prior notification to the qualifying activity.

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APPENDIX

TABLE XVI. Critical resistance values for qualification ranges. 1/

Style	Slash sheet	Critical resistance value 2/							
		Characteristic 3/							
		E		H		K		M	
		Ceramic board	FR-4 board	Ceramic board	FR-4 board	Ceramic board	FR-4 board	Ceramic board	FR-4 board
RM0502	/1	158.0	158.0	158.0	158.0	78.7	78.7	78.7	78.7
RM0505	/2	63.4	63.4	63.4	63.4	31.6	31.6	31.6	31.6
RM0705	/6	49.9	49.9	49.9	49.9	24.9	24.9	24.9	24.9
RM1005	/3	31.6	31.6	31.6	31.6	15.8	15.8	15.8	15.8
RM1010	/10	6.34	6.34	6.34	6.34	3.16	3.92	3.16	3.92
RM1505	/4	15.8	15.8	15.8	15.8	10.6	10.6	10.6	10.6
RM2208	/5	7.87	7.87	7.87	7.87	6.98	6.98	6.98	6.98
RM1206	/7	78.7	78.7	78.7	78.7	39.2	39.2	39.2	39.2
RM2010	/8	54.9	54.9	54.9	54.9	28.0	37.4	28.0	37.4
RM2512	/9	78.7	78.7	78.7	78.7	39.2	52.3	39.2	52.3

1/ Maximum continuous working voltage shall be applied (see 3.1).
2/ The critical resistance value is the maximum standard resistance value which will dissipate full wattage when the maximum continuous working voltage is applied.
3/ All resistance in kilohms.

40. EXTENT OF QUALIFICATION

40.1 Extent of qualification. The resistance range included in the qualification of any one style and tolerance shall be between any two adjacent resistance values which pass the required qualification inspection. The extent of qualification between styles, characteristics, terminations, failure rate levels, and resistance tolerances shall be as specified in table XVII.

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APPENDIX

TABLE XVII. Extension of qualification.

Style	Will qualify style	
RM0502	RM0502	
RM0505	RM0505	
RM0705	RM0705, RM0505	
RM1005	RM1005, RM0705, RM0505	
RM1010	RM1010, RM1005, RM0705, RM0505	
RM1505	RM1505, RM1005, RM0705, RM0505	
RM2208	RM2208	
RM1206	RM1206, RM1505, RM1005, RM0705, RM0505	
RM2010	RM2208, RM2010	
RM2512	RM2010, RM2208, RM2512	
Characteristic	**Will qualify characteristic**	**May qualify characteristic with minor additional testing 1/**
E	E, H	K, M
H	H	K, M
K	K, M	----
M	M	----
Termination	**Will qualify termination**	**May qualify termination with minor additional testing 1/**
G	G, W, P	B, R, S
W	W, P	S
P	P, W	S
B	B, R, S	G, W, P 2/
R	R, S	-------
S	S	-------
U	U, T	C, D
T	T	D
C	C, D	U, T
D	D	T
Failure rate level	**Will qualify failure rate level**	
S	S, R, P, M	
R	R, P, M	
P	P, M	
M	M	
Resistance tolerance	**Will qualify resistance tolerance**	
0.1	0.1, 1.0, 2.0, 5.0, 10.0	
1.0	1.0, 2.0, 5.0, 10.0	
2.0	2.0, 5.0, 10.0	
5.0	5.0, 10.0	
10.0	10.0	

1/ To be determined by qualifying activity.
2/ Must contain barrier layer to qualify.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, AT, AV, ME
Navy - AS, CG, MC, OS
Air Force - 17, 19, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5905-1291)

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-55342F	2. DOCUMENT DATE (YYMMDD) 95/06/28

3. DOCUMENT TITLE
RESISTORS, FIXED, FILM, CHIP, ESTABLISHED RELIABILITY, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Research Laboratory	b. TELEPHONE (Include Area Code) (1) Commercial (908) 427-3441 (2) AUTOVON 987-3441
c. ADDRESS (Include Zip Code) Physicals Sciences Directorate ATTN: AMSRL-PS-DC Fort Monmouth, NJ 07703-5601	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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